Wireshark Beginner Guide

Practical Packet Analysis for Cybersecurity & Networking

A beginner-focused practical handbook covering packet analysis, filters, troubleshooting, cybersecurity investigations, and network traffic analysis using Wireshark.

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1. What is Wireshark?

Wireshark is a network protocol analyzer used to inspect and analyze network traffic in real time. It allows cybersecurity professionals, network engineers, and IT administrators to view packets traveling across a network.

Wireshark helps you understand:
• Which devices are communicating
• Which protocols are being used
• Whether suspicious traffic exists
• Why a network problem is happening

Basic Packet Flow

Quick Tip: Wireshark does not hack networks by itself. It is mainly a visibility and analysis tool.

2. Installing Wireshark

Wireshark is available for Windows, Linux, and macOS.

Operating System	Installation Method
Windows	Download installer and install Npcap
Linux	sudo apt install wireshark
macOS	Install using dmg or Homebrew

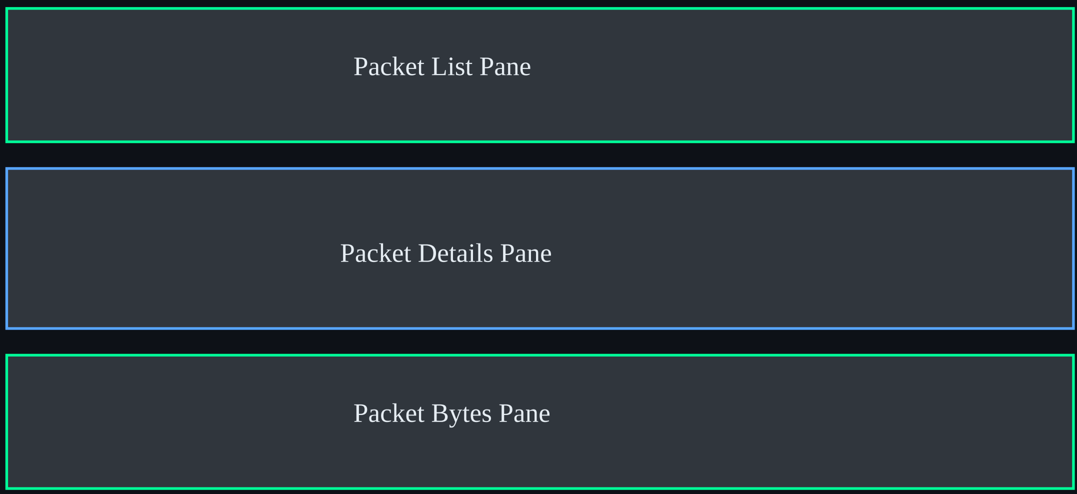
During installation on Windows, ensure Npcap is installed because Wireshark requires it to capture packets.

Quick Tip: Run Wireshark as administrator when troubleshooting interface or permission issues.

3. Understanding the Interface

The Wireshark interface is divided into three major sections:

Packet List Pane — Displays captured packets
Packet Details Pane — Shows protocol details
Packet Bytes Pane — Displays raw hexadecimal packet data



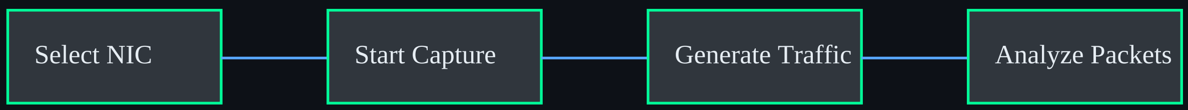
Quick Tip: Start learning by focusing on source IPs, destination IPs, protocols, and ports before diving into raw bytes.

4. Capturing Packets

Packet capture means recording network traffic passing through a network interface.

Steps:
1. Open Wireshark
2. Select a network interface
3. Click Start Capture
4. Generate traffic by browsing websites or pinging hosts
5. Stop the capture

Packet Capture Process



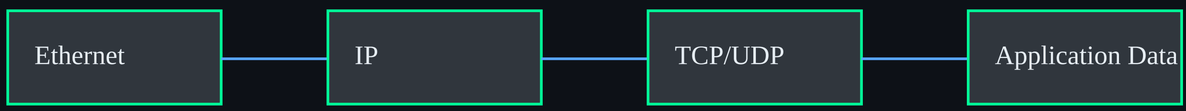
Common interfaces:
• Ethernet
• Wi■Fi
• Virtual adapters
• VPN interfaces

Quick Tip: Avoid capturing on every interface at once unless necessary. It creates noise and large capture files.

5. Packet Structure Basics

Each packet contains multiple layers. Wireshark separates these layers to help analysts inspect network communication.

Packet Layer Breakdown



Ethernet Layer: Hardware communication using MAC addresses
IP Layer: Logical routing using IP addresses
TCP/UDP Layer: Transport communication
Application Layer: HTTP, DNS, FTP, etc.

Example:
Source IP: 192.168.1.10
Destination IP: 142.250.190.14
Protocol: HTTPS
Port: 443

Quick Tip: Understanding layers makes troubleshooting easier because problems often happen at a specific layer.

6. Wireshark Filters

Filters allow analysts to isolate relevant traffic quickly.

Filter	Purpose
http	Show HTTP traffic
dns	Show DNS packets
ip.addr == 192.168.1.5	Traffic from specific IP
tcp.port == 443	HTTPS traffic
icmp	Ping traffic
arp	ARP packets

Display filters reduce visual noise and help investigators focus on important packets.

Quick Tip: Mastering filters is one of the fastest ways to become efficient with Wireshark.

7. Protocol Analysis

HTTP
Web browsing traffic

DNS
Domain name lookups

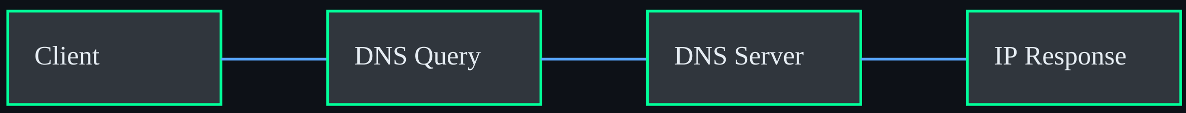
TCP
Reliable communication

UDP
Fast lightweight communication

ICMP
Ping and diagnostics

ARP
Local network device discovery

DNS Request Flow



TCP Handshake

Wireshark can reconstruct sessions and show request/response behavior, making it useful for protocol learning and troubleshooting.

8. Cybersecurity Use Cases

Wireshark is heavily used in cybersecurity operations.

Use Case	Example
Malware Analysis	Suspicious outbound traffic
Intrusion Detection	Unknown IP communication
Credential Theft	Plain-text protocols
DNS Analysis	Malicious domain lookups
Reconnaissance	Port scanning activity

Example investigation:
1. Identify suspicious IP
2. Filter traffic
3. Analyze protocols used
4. Check unusual destinations
5. Export packets for investigation

Quick Tip: Unencrypted protocols like Telnet and FTP expose credentials in plain text.

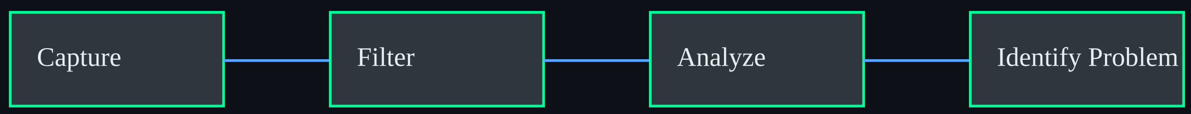
9. Troubleshooting Scenarios

Scenario 1: Website not loading
• Use DNS filters to check resolution
• Inspect TCP handshake
• Verify HTTPS traffic

Scenario 2: Slow network
• Identify high traffic sources
• Check retransmissions
• Analyze latency

Scenario 3: Connectivity failure
• Use ICMP packets
• Check ARP requests
• Verify gateway communication

Troubleshooting Flow



10. Best Practices

• Capture only necessary traffic

• Use filters aggressively

• Save important captures

• Document investigation findings

• Learn common protocols deeply

• Avoid analyzing production traffic carelessly

Quick Tip: Large captures become difficult to analyze. Small focused captures are usually more effective.

11. Practice Labs & Exercises

Beginner Practice Labs:

• Capture your own web browsing traffic
• Analyze DNS queries
• Inspect TCP handshakes
• Identify devices on your network
• Capture ping traffic
• Detect a port scan using Nmap

Exercise	Goal
Filter only DNS traffic	Learn DNS behavior
Identify your IP address	Understand endpoints
Find HTTPS packets	Recognize encrypted traffic
Capture ICMP packets	Understand ping requests

Wireshark becomes easier when combined with hands-on practice. Build a small home lab and capture traffic regularly.

Quick Tip: The goal is not just seeing packets — it is understanding what normal and abnormal traffic looks like.